12.1 Curve in Space and Their Tangents

Vector Functions

Recall that a line in space is represented by the vector equation

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}.$$

The equation can also be written as

$$\mathbf{r}(t) = (a + tv_1)\mathbf{i} + (b + tv_2)\mathbf{j} + (c + tv_3)\mathbf{k},$$

Where $\mathbf{r}_0 = \langle a, b, c \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$.

This is called a **vector function** (or **vector-valued function**). It is a function that takes variables and returns a vector. We will look at vector functions other than lines.

Note: For this section, we will only look at vector functions of a single variable (which generate curves). Vector functions of two variables shall be considered in a later course.

A vector function is often used to represent a moving particle's position in the space during a time interval I:

$$x = f(t),\ \ y = g(t),\ \ z = h(t),\ \ \ t \in I.$$

The points $(x, y, z) = (f(t), g(t), h(t)),\ t \in I$ make up the **curve** in space - the particle's **path**. The particle's **position** at time t is then $P(f(t), g(t), h(t))$.

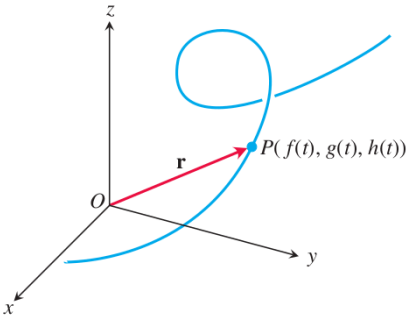


FIGURE 12.1 The position vector $\mathbf{r} = \overrightarrow{OP}$ of a particle moving through space is a function of time.

In general, a vector function of a single variable in $\mathbb{R}^2$ and $\mathbb{R}^3$ has the form

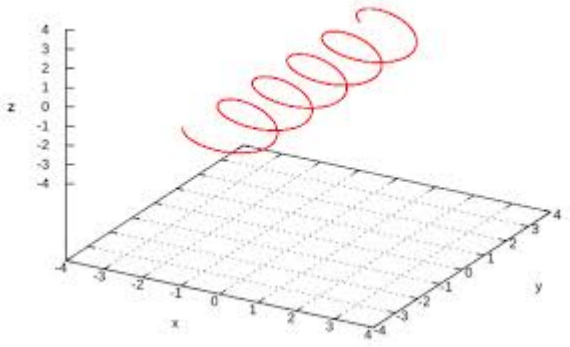
$$\mathbf{r}(t) = \langle f(t), g(t) \rangle \quad \& \quad \mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$$

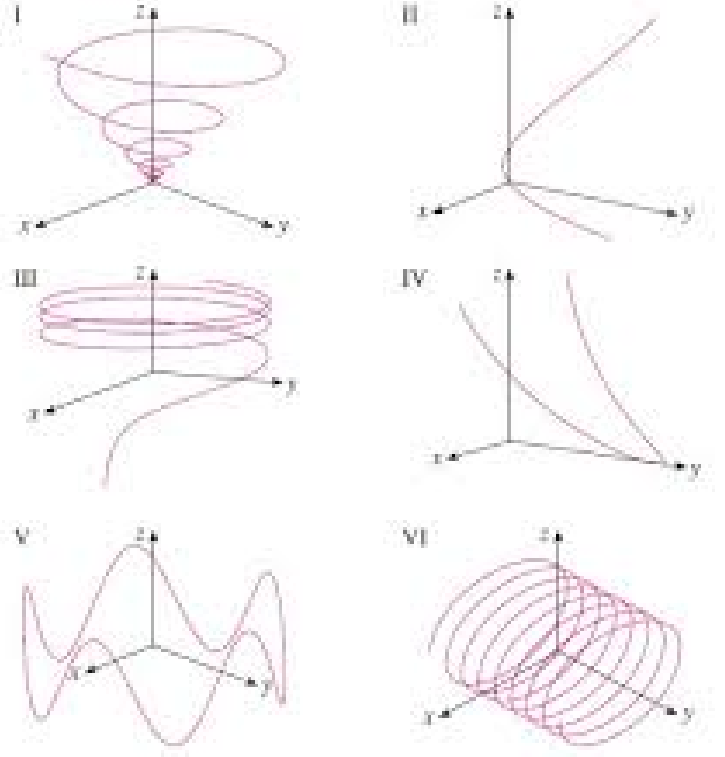
OR

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} \ \ \& \ \ \mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

respectively, where $f(t), g(t), h(t)$ are called the **component functions**.

Example of vector functions of single variables:





When considering a general vector function, the domain of the vector function shall be determined first. The **domain** of a vector function is the set of all t's for which all the component functions are defined.

Example: Consider the following vector function.

$$\mathbf{r}(t) = \langle 2\sin t, \sqrt{16 - t^2}, \ln(5 - 2t) \rangle$$

Example: (Graphing a vector function in $\mathbb{R}^2$)

Graph the vector function $\mathbf{r}(t) = 2t\mathbf{i} + (t^2 + 1)\mathbf{j}$.

Example: (A vector function in $\mathbb{R}^3$)

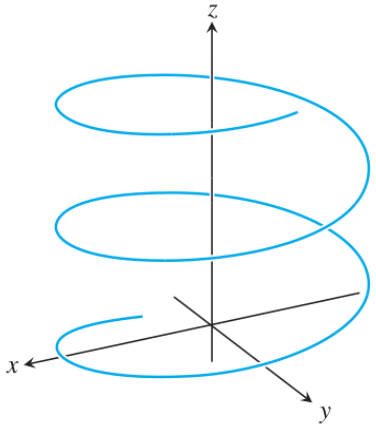
Observe that the sketch of the following vector function

$$\mathbf{r}(t) = \langle 2 - 4t, -1 + 5t, 2 + t \rangle$$

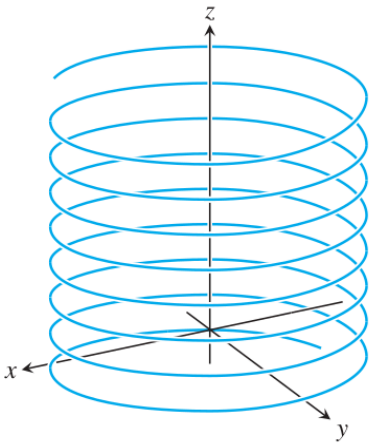
is a line in space.

Example: (Graphing a vector function in $\mathbb{R}^3$)

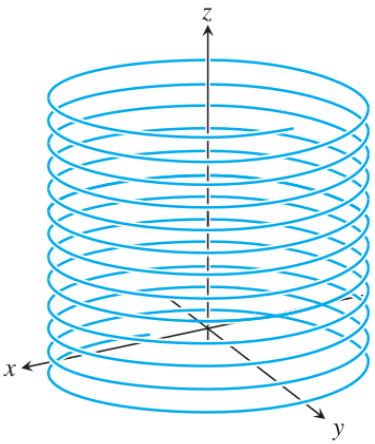
Graph the vector function $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$.



$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$



$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + 0.3t\mathbf{k}$



$\mathbf{r}(t) = (\cos 5t)\mathbf{i} + (\sin 5t)\mathbf{j} + t\mathbf{k}$

The curve in the previous example is called **helix** (plural - **helixes**).

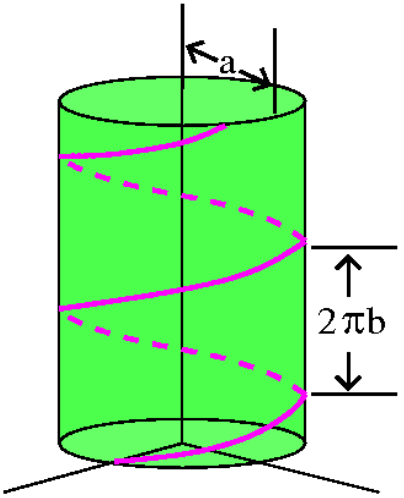
(Note: *It has the property that the tangent line at any point makes a constant angle with a fixed line.*)

The helix in the previous example is defined by the following parametrization.

$$x = \cos t$$
$$y = \sin t$$
$$z = t$$

A general circular helix has the following parametrization.

$$x = a\cos t$$
$$y = a\sin t$$
$$z = bt$$



Limits and Continuity

Definition: Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ be a vector function with domain D, and let $\mathbf{L}$ be a vector. We say that $\mathbf{r}$ has **limit** $\mathbf{L}$ as t approaches t_0 and write

$$\lim_{t \to t_0} \mathbf{r}(t) = \mathbf{L}$$

if, for every $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all $t \in D$

$$|\mathbf{r}(t) - \mathbf{L}| < \epsilon \text{ whenever } 0 < |t - t_0| < \delta.$$

Practically, the limit of a vector function can be view as

$$\begin{aligned}
\lim_{t \to t_0} \mathbf{r}(t) &= \lim_{t \to t_0} \langle f(t), g(t), h(t) \rangle \\
&= \langle \lim_{t \to t_0} f(t), \lim_{t \to t_0} g(t), \lim_{t \to t_0} h(t) \rangle \\
&= \left(\lim_{t \to t_0} f(t) \right) \mathbf{i} + \left(\lim_{t \to t_0} g(t) \right) \mathbf{j} + \left(\lim_{t \to t_0} h(t) \right) \mathbf{k}
\end{aligned}$$

Example: If $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$, then

$$\lim_{t\to\pi/4} \mathbf{r}(t) =$$

Example: If $\mathbf{r}(t) = \langle t^3, \frac{\sin(3t-3)}{t-1}, e^{2t}\rangle$, then

$$\lim_{t\to 1} \mathbf{r}(t) =$$

Definition: A vector function $\mathbf{r}(t)$ is **continuous at a point** $t = t_0$ in its domain if $\lim_{t \to t_0} \mathbf{r}(t) = \mathbf{r}(t_0)$. The function is **continuous** if it is continuous at every point in its domain.

Remark: From the definition of limit, it is easy to see that a vector function $\mathbf{r}(t)$ is continuous at $t = t_0$ if and only if each component function is continuous there.

For example, the vector function

$$\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$$

is continuous for all $t \in \mathbb{R}$.

On the other hand, the vector function

$$\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + \lfloor t \rfloor \mathbf{k}$$

is discontinuous at every integer t.

Derivatives and Motion

Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ be the position vector of a moving particle. Suppose that f, g, h are differentiable. Define $\Delta\mathbf{r}$ to be the difference between the particle's position at time t and time $t + \Delta t$.

Definition:

The vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ has a **derivative (is differentiable) at** t if f, g, and h have derivatives at t. The derivative is the vector function

$$\mathbf{r}'(t) = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \frac{df}{dt}\mathbf{i} + \frac{dg}{dt}\mathbf{j} + \frac{dh}{dt}\mathbf{k} = f'\mathbf{i} + g'\mathbf{j} + h'\mathbf{k}$$

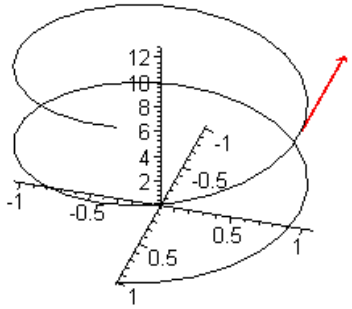
Example:

The derivative of $\mathbf{r}(t) = t^6\mathbf{i} + \sin(2t)\mathbf{j} - \ln(t+1)\mathbf{k}$ is

We say a vector function $\mathbf{r}$ is **differentiable** if it is differentiable at every point of its domain.

A **smooth curve** is a curve $\mathbf{r}$ for which $\frac{d\mathbf{r}}{dt}$ is continuous and never $\mathbf{0}$. For example, a helix is a smooth curve.

Tangent and tangent line of a curve



On a smooth curve, if the point *P* has position vector $\mathbf{r}(t)$, then the vector $\mathbf{r}'(t)$ is defined to be the vector **tangent** to the curve at *P*.

The **tangent line** to the curve at a point $\mathbf{r}(t_0)$ is defined to be the line through the point parallel to $\mathbf{r}'(t_0)$.

Example: Let $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ be a smooth curve. Find the equation of the tangent line at $t = \frac{\pi}{4}$.

Definition: Let $\mathbf{r}$ is the position vector of a particle moving along a smooth curve in space.

1. Velocity is the derivative to position:
$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}.$$

2. Speed is the magnitude of velocity:
$$\text{Speed} = |\mathbf{v}|.$$

3. Acceleration is the derivative of velocity:
$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}.$$

4. The unit vector $\mathbf{v}/|\mathbf{v}|$ is the direction of motion at time t.

The velocity of a moving particle can be expressed as the product of its speed and direction:
$$\text{Velocity} = |\mathbf{v}| \left(\frac{\mathbf{v}}{|\mathbf{v}|} \right) = \text{(speed)(direction)}.$$

Example: Find the velocity, speed, acceleration of a particle whose motion in space is given by position vector $\mathbf{r}(t) = (2\cos t)\mathbf{i} + (2\sin t)\mathbf{j} + (5\cos^2 t)\mathbf{k}$.

Example: A particle's motion in space is given by position vector $\mathbf{r}(t) = (2\cos t)\mathbf{i} + (2\sin t)\mathbf{j} + (5\cos^2 t)\mathbf{k}$. Find the times, if any, when the particle's acceleration is orthogonal to its velocity.

Differentiation Rules for Vector Functions

Let $\mathbf{u}$ and $\mathbf{v}$ be differentiable vector functions of t, $\mathbf{C}$ a constant vector, c any scalar, and f any differentiable scalar function of t.

1. Constant Function Rule: $\dfrac{d}{dt}\mathbf{C} = \mathbf{0}$
2. Scalar Multiple Rules: $\dfrac{d}{dt}\left[c\mathbf{u}(t)\right] = c\mathbf{u}'(t)$ & $\dfrac{d}{dt}\left[f(t)\mathbf{u}(t)\right] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
3. Sum Rule: $\dfrac{d}{dt}\left[\mathbf{u}(t) + \mathbf{v}(t)\right] = \mathbf{u}'(t) + \mathbf{v}'(t)$
4. Difference Rule: $\dfrac{d}{dt}\left[\mathbf{u}(t) - \mathbf{v}(t)\right] = \mathbf{u}'(t) - \mathbf{v}'(t)$
5. Dot Product Rule: $\dfrac{d}{dt}\left[\mathbf{u}(t) \cdot \mathbf{v}(t)\right] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$
6. Cross Product Rule: $\dfrac{d}{dt}\left[\mathbf{u}(t) \times \mathbf{v}(t)\right] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$
7. Chain Rule: $\dfrac{d}{dt}\left[\mathbf{u}(f(t))\right] = f'(t)\mathbf{u}'(f(t))$

Proof of the Dot Product Rule:

$$\frac{d}{dt}\left[\mathbf{u}(t)\cdot\mathbf{v}(t)\right]=\mathbf{u}'(t)\cdot\mathbf{v}(t)+\mathbf{u}(t)\cdot\mathbf{v}'(t)$$

Proof of the Cross Product Rule:

$$\frac{d}{dt}\left[\mathbf{u}(t) \times \mathbf{v}(t)\right] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$$

Proof of the Chain Rule:

$$\frac{d}{dt}\left[\mathbf{u}(f(t))\right] = f'(t)\mathbf{u}'(f(t))$$

Vector Functions of Constant Length

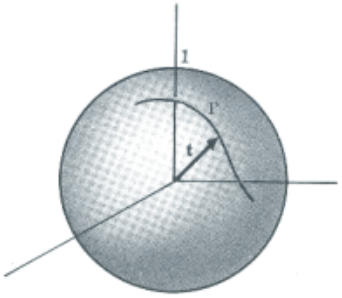


Fig. 3

When a particle move on a sphere centered at the origin, its position vector $\mathbf{r}(t)$ has a constant length equal to the radius of the sphere.

The velocity vector $d\mathbf{r}/dt$, tangent to the motion, is tangent to the sphere and hence perpendicular to $\mathbf{r}$. This is always the case for a differentiable vector function of constant length: $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal.

From direct calculation:

Observation: If $\mathbf{r}$ is a differentiable vector function of t of constant length, then

$$\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = 0.$$

Example: Show that $\mathbf{r}(t) = (\sin t)\mathbf{i} + (\cos t)\mathbf{j} + \sqrt{3}\mathbf{k}$ has constant length and is orthogonal to its derivative.

12.2 Integrals of Vector Functions

Definition:

The **indefinite integral** of $\mathbf{r}$ with respect to t is the set of all antiderivative of $\mathbf{r}$, denoted by $\int \mathbf{r}(t)\ dt$. If $\mathbf{R}$ is any antiderivative of $\mathbf{r}$, then

$$\int \mathbf{r}(t)\ dt = \mathbf{R}(t) + \mathbf{C}.$$

Practically,

$$\int \mathbf{r}(t)\ dt = \langle \int f(t)\ dt, \int g(t)\ dt, \int h(t)\ dt \rangle + \mathbf{C},$$

$$\int \mathbf{r}(t)\ dt = \left(\int f(t)\ dt\right)\mathbf{i} + \left(\int g(t)\ dt\right)\mathbf{j} + \left(\int h(t)\ dt\right)\mathbf{k} + \mathbf{C}.$$

Example:

$$\int \left((\cos t)\mathbf{i} + \mathbf{j} - 2t\mathbf{k}\right)\ dt =$$

Definition:

If the components of $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ are integrable over $[a, b]$, then so is $\mathbf{r}$, and the **definite integral** of $\mathbf{r}$ from a to b is

$$\int_a^b \mathbf{r}(t)\ dt = \left(\int_a^b f(t)\ dt\right)\mathbf{i} + \left(\int_a^b g(t)\ dt\right)\mathbf{j} + \left(\int_a^b h(t)\ dt\right)\mathbf{k}.$$

The Fundamental Theorem of Calculus for continuous vector functions:

$$\int_a^b \mathbf{r}(t)\ dt = \mathbf{R}(t)\Big]_a^b = \mathbf{R}(b) - \mathbf{R}(a),$$

where $\mathbf{R}$ is any antiderivative of $\mathbf{r}$.

Example:

$$\int_0^{\pi} ((\cos t)\mathbf{i} + \mathbf{j} - 2t\mathbf{k})\ \ dt =$$

Example: Find the position vector $\mathbf{r}(t)$ of a particle if its velocity is $\mathbf{v}(t) = 2t\mathbf{i} + (\sin t)\mathbf{j} + (\cos t)\mathbf{k}$ and its initial position is $\mathbf{r}(0) = \mathbf{i} + \mathbf{j} + \mathbf{k}$.

Example: Suppose we do not know the path of a hang glider, but only its acceleration vector $\mathbf{a}(t) = -(3\cos t)\mathbf{i} - (3\sin t)\mathbf{j} + 2\mathbf{k}$. We also know that initially (at time $t = 0$) the glider departed from the point $(4, 0, 0)$ with velocity $\mathbf{v}(0) = 3\mathbf{j}$. Find the glider's position as a function of t.

